

Day 17 Quiz

Geometry Summer Credit | Central Angles, Inscribed Angles, and Arcs

Name:	Date:	Score:
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Directions

Choose the best answer. Show the angle-arc relationship for short-response items.

1.	A central angle has its vertex: A outside the circle B on the circle C at the center D at the midpoint of a chord
2.	A central angle measuring 82 degrees intercepts an arc measuring: A 41 degrees B 82 degrees C 164 degrees D 278 degrees
3.	An inscribed angle intercepts a 150-degree arc. The angle measures: A 75 degrees B 150 degrees C 210 degrees D 300 degrees
4.	An inscribed angle measures 48 degrees. Its intercepted arc measures: A 24 degrees B 48 degrees C 96 degrees D 312 degrees
5.	An inscribed angle intercepting a diameter measures: A 45 degrees B 90 degrees C 180 degrees D 360 degrees
6.	Minor arc AB measures 125 degrees. Find major arc AB.
7.	Three central angles are x , $2x$, and $5x$. Find x and all three arc measures.

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Continued

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| 8. | An inscribed angle is $4x+3$ degrees and its intercepted arc is $10x-6$ degrees. Find x and the angle. |
| 9. | A central angle is $6x+8$ degrees and an inscribed angle intercepting the same arc is $2x+24$ degrees. Find x and both angles. |
| 10. | Spiral review: Write the equation of a circle centered at $(-3,2)$ with radius 5. |
| 11. | Error analysis: A student says a 68-degree inscribed angle intercepts a 34-degree arc. Explain and correct both the relationship and arc measure. |